The Effects of Airfoil Geometry on Lift-to-Drag Performance

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Abstract

The lift-to-drag ratio (L/D) sets how far an aircraft can glide and how long a drone can fly on one battery charge. This study used the XFOIL panel code to examine how camber and thickness affect the maximum lift-to-drag ratio, (L/D)max, of five NACA 4-digit airfoils (0012, 2412, 4412, 0015, and 0018) at Reynolds numbers of 100,000 to 1,000,000. Angle-of-attack sweeps from −5° to 15° were automated in Python, and 98% of the 820 points converged. Validation against NACA Report 824 wind-tunnel data for the NACA 2412 at Re = 3,100,000 showed that XFOIL over-predicted maximum lift by about 11% and under-predicted minimum drag by about 16%. Camber had the largest effect: at Re = 1,000,000, (L/D)max rose from 75.6 for the NACA 0012 to 132.9 for the NACA 4412. Contrary to the hypothesis, increasing thickness from 12% to 18% did not reduce (L/D)max, which changed by less than 8%. (L/D)max increased with Reynolds number for every airfoil, with the largest gains between 100,000 and 500,000. These results suggest that a moderately cambered section such as the NACA 4412 offers the highest efficiency for small aircraft operating at low Reynolds numbers.

Introduction

Airfoil efficiency is important because it affects how far an aircraft can fly, how much fuel it uses, and how long a drone can stay in the air on one battery charge. The lift-to-drag ratio (L/D) measures how efficiently an airfoil produces lift compared to the drag it creates. Choosing the right airfoil can be challenging because different shapes perform differently depending on flight conditions, especially Reynolds number, and small drones fly at low Reynolds numbers where these differences are largest. This study uses XFOIL simulations to measure how camber and thickness affect the maximum L/D of five NACA airfoils across angles of attack and Reynolds numbers from 100,000 to 1,000,000.

Background and Literature Review

Aerodynamic fundamentals

Lift is the force that allows an aircraft to stay in the air; it acts perpendicular to the oncoming airflow. Drag is the air resistance that acts parallel to the airflow, against the aircraft's forward movement. The lift-to-drag ratio (L/D) compares these two forces to show how efficiently an airfoil performs. The angle of attack is the angle between the airfoil's chord line and the oncoming airflow, and changing it affects how much lift and drag are produced. Reynolds number describes how airflow behaves around an airfoil based on airspeed, air density, viscosity, and airfoil size; it compares inertial forces to viscous forces. It is important because different airfoils can perform differently at high and low Reynolds numbers, especially on smaller aircraft and drones.

Reynolds number is defined as

$$Re = \frac{\rho V c}{\mu}$$

where ρ is air density (kg/m³), V is airspeed (m/s), c is chord length (m), and μ is the dynamic viscosity of air (kg/(m·s)). Lift and drag are compared as dimensionless coefficients per unit span, so that airfoils of different sizes and speeds can be compared fairly:

$$C_l = \frac{L'}{\tfrac{1}{2}\rho V^2 c} \qquad C_d = \frac{D'}{\tfrac{1}{2}\rho V^2 c} \qquad \frac{L}{D} = \frac{C_l}{C_d}$$

Prior work

The NACA 4-digit airfoils were tested extensively in the NACA two-dimensional low-turbulence wind tunnel. Abbott, von Doenhoff, and Stivers (1945) compiled lift, drag, and moment data for these sections at Reynolds numbers from 3 to 9 million, and Abbott and von Doenhoff (1959) later published the same data with the underlying theory. These measurements remain a standard reference for checking computational methods. However, they do not cover the Reynolds numbers below 1 million at which small drones and model aircraft fly.

At low Reynolds numbers, the boundary layer stays laminar over more of the surface and can separate and reattach, forming a laminar separation bubble that raises drag and changes the lift curve (Lissaman, 1983; Mueller & DeLaurier, 2003). XFOIL (Drela, 1989) was designed for this regime. It couples a panel method for the outer flow with an integral

boundary-layer model and predicts transition with the e^N method, where the parameter Ncrit sets how quickly disturbances trigger turbulence. XFOIL is fast and widely used, but it is known to be less accurate once large regions of flow separate, as happens near stall (Drela, 1989).

Research Question and Hypotheses

Research question: How do airfoil camber and thickness affect the maximum lift-to-drag ratio, (L/D)max, and the angle of attack at which it occurs, across Reynolds numbers from 100,000 to 1,000,000?

- **H1 (camber):** Increasing camber raises (L/D)max and lowers the angle of attack at which it occurs, because a cambered section produces lift at lower angles of attack.
- **H2 (thickness):** Increasing thickness lowers (L/D)max, because a thicker section produces more pressure drag.
- **H3 (Reynolds number):** (L/D)max increases with Reynolds number for all airfoils, with the largest gain between 100,000 and 500,000, because skin-friction drag falls as Reynolds number rises.

Methodology

Five NACA 4-digit airfoils were simulated in XFOIL 6.99 (Drela, 1989) at four Reynolds numbers, giving 20 angle-of-attack sweeps plus one validation run.

Airfoil selection

The airfoils form two series that share the NACA 0012 as a baseline, so that only one geometric feature changes within each series.

Series	Airfoils	Variable changed	Held constant
Camber	NACA 0012, 2412, 4412	Maximum camber: 0%, 2%, 4% of chord	Thickness: 12% of chord
Thickness	NACA 0012, 0015, 0018	Maximum thickness: 12%, 15%, 18% of chord	Camber: 0%

All coordinates were generated with XFOIL's built-in NACA 4-digit generator, so every shape came from the same analytical definition.

Variables

- **Independent:** camber, thickness, Reynolds number (100,000; 200,000; 500,000; 1,000,000), angle of attack α.
- **Dependent:** lift coefficient Cl, drag coefficient Cd, L/D, (L/D)max, and α at (L/D)max.
- **Controlled:** Mach number 0 (incompressible flow); transition parameter Ncrit = 9 (a clean wind tunnel or calm air); 160 panel nodes; 100 viscous iterations per point; α step of 0.5°.

Simulation procedure

1. Generate the airfoil with the `NACA` command and set 160 panel nodes (`PPAR`).
2. Enter viscous mode at the target Reynolds number (`VISC`), and set Mach = 0 and Ncrit = 9.
3. Sweep α from 0° to 15°, reset the boundary-layer solution (`INIT`), then sweep from 0° to −5°. Splitting the sweep at 0° reduces convergence failures at negative angles.
4. Save each sweep as a polar file named by airfoil and Reynolds number.
5. Repeat for all 5 airfoils × 4 Reynolds numbers (20 sweeps).

The runs were automated with a Python script that sent the same XFOIL commands for every case, deleted any old polar file before each run, removed duplicate rows, and logged angles that failed to converge.

Data processing

L/D was computed as Cl/Cd at every converged point. For each airfoil and Reynolds number, (L/D)max and its angle of attack were taken as the largest L/D in the sweep. With a 0.5° step, the angle of (L/D)max is resolved to ±0.25°. Non-converged points were reported rather than removed or filled in.

Validation

The NACA 2412 was simulated at Re = 3,100,000, from −5° to 20°, and compared with wind-tunnel data from NACA Report 824, page 136 (Abbott, von Doenhoff, & Stivers, 1945). This is the lowest Reynolds number in that report. The comparison used maximum lift coefficient, the angle at maximum lift, and minimum drag coefficient.

Software

XFOIL 6.99 was installed on macOS 15 (Apple M4) through a Homebrew package patched for Apple Silicon. XFOIL's graphics were disabled, and all figures were produced in Python 3.9 with matplotlib.

Results

The NACA 4412 reached the highest (L/D)max at every Reynolds number, peaking at 132.9 at Re = 1,000,000. Of the 820 points in the 20 main sweeps, 804 converged (98.0%). The 16 missing points were isolated single angles, and none fell at or next to an (L/D)max peak.

Maximum lift-to-drag ratio

Table 1. (L/D)max, with the angle of attack at which it occurs in parentheses.

Airfoil	Re = 100,000	Re = 200,000	Re = 500,000	Re = 1,000,000
NACA 0012	36.7 (5.0°)	47.4 (5.0°)	61.8 (6.0°)	75.6 (7.5°)
NACA 2412	50.2 (7.0°)	67.3 (6.5°)	89.3 (5.5°)	104.7 (4.5°)
NACA 4412	55.4 (9.0°)	78.1 (7.5°)	108.6 (7.0°)	132.9 (5.5°)
NACA 0015	37.4 (6.0°)	49.5 (6.0°)	66.5 (7.5°)	78.9 (9.0°)
NACA 0018	37.5 (6.5°)	50.1 (7.0°)	65.7 (9.0°)	78.0 (10.0°)

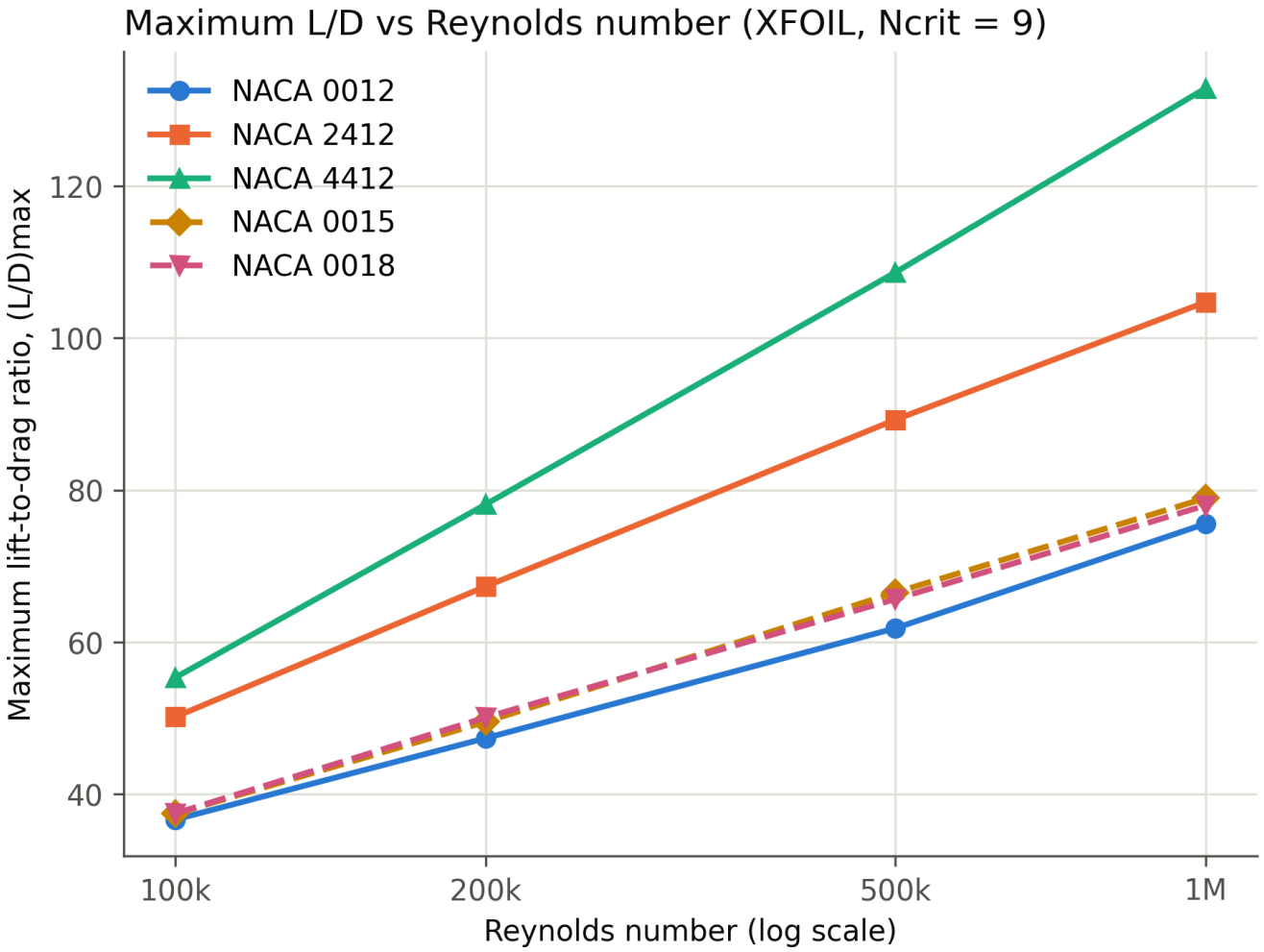


Figure 4. (L/D)max versus Reynolds number. Solid lines: camber series. Dashed lines: thickness series.

Lift, drag, and L/D curves

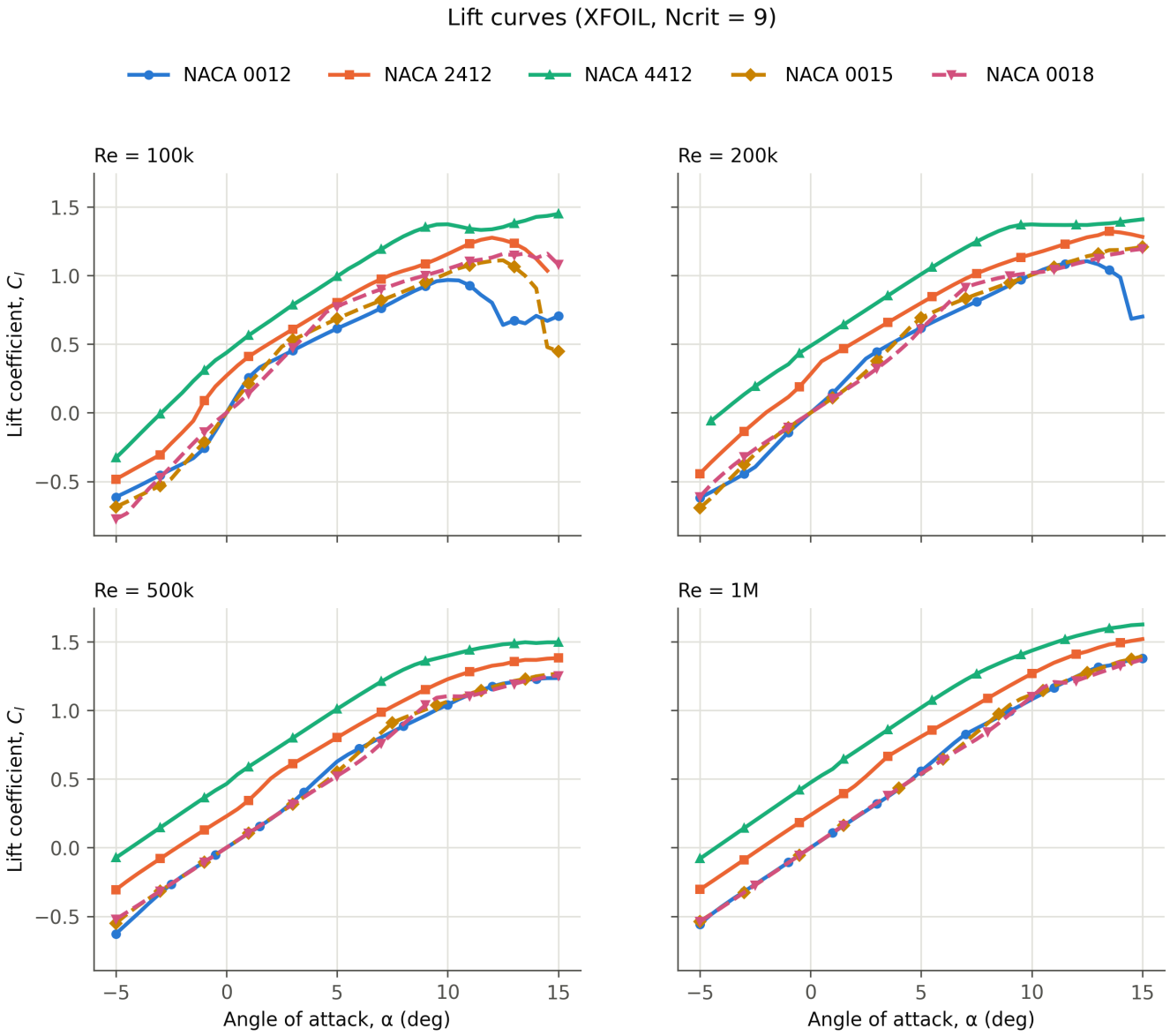


Figure 1. Lift curves. At Re = 100,000 the NACA 0012 stalls near 10°, while the NACA 0018 continues to gain lift until about 14°. At Re = 500,000 and above, no airfoil reached a clear stall within the 15° sweep.

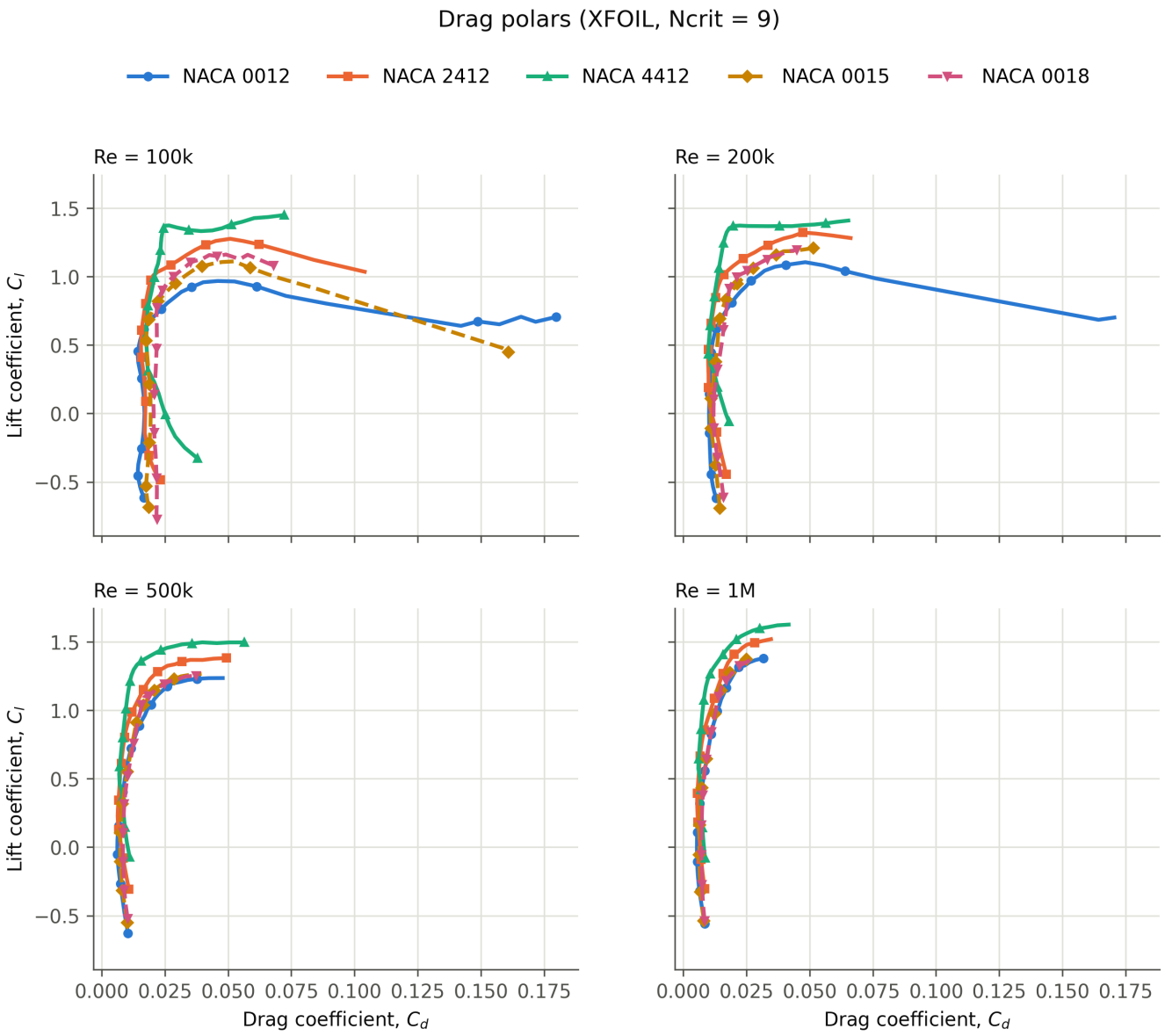


Figure 2. Drag polars (Cl versus Cd).

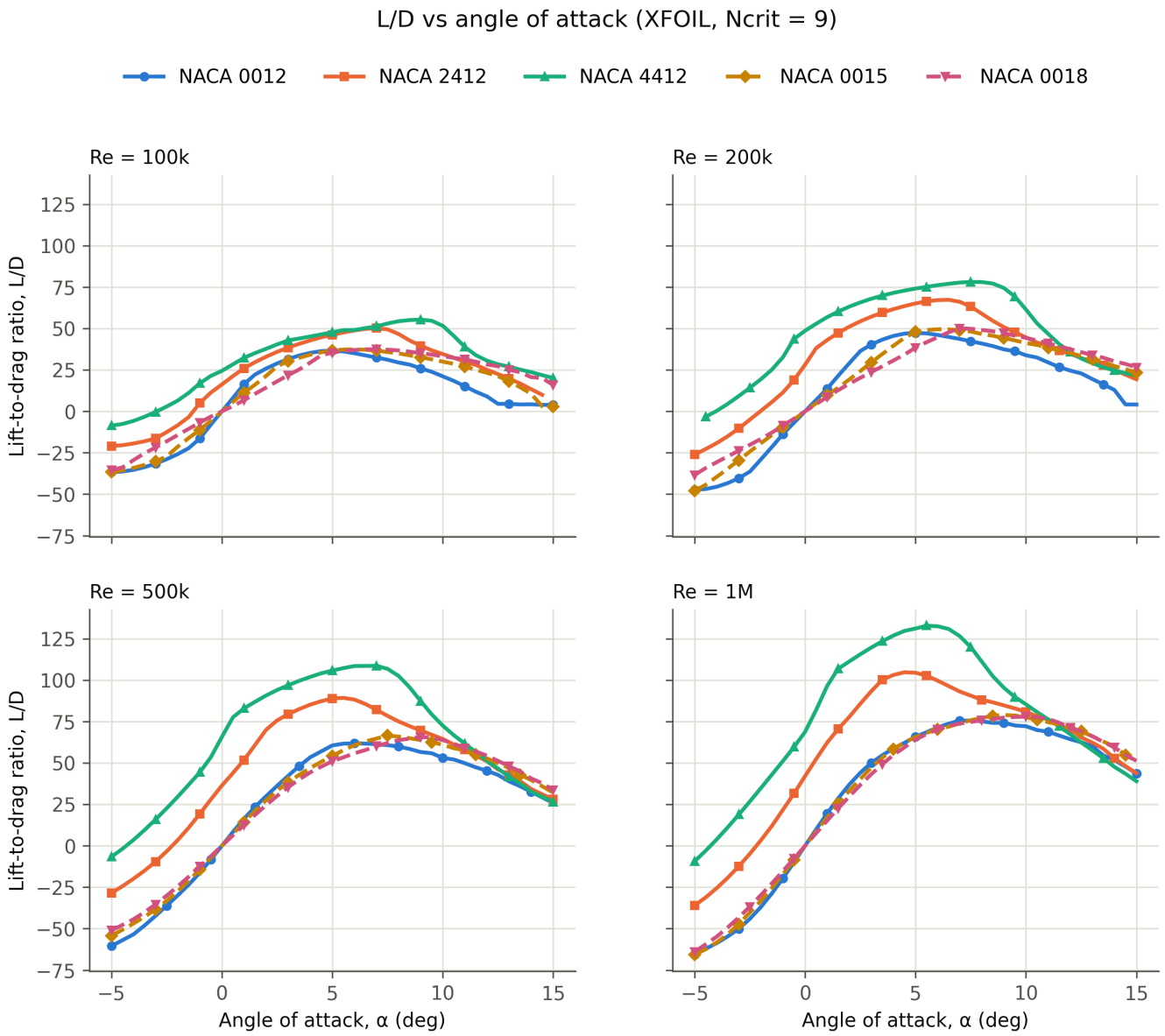


Figure 3. L/D versus angle of attack.

Validation

Table 2. XFOIL compared with NACA Report 824 wind-tunnel data for the NACA 2412 at Re = 3,100,000. All 51 validation points converged.

Quantity	NACA Report 824	XFOIL	Difference
Maximum Cl	1.6	1.77	+10.6%
α at maximum Cl	about 15°	18.5°	+3.5°
Minimum Cd	0.006	0.00507	−15.5%

Report values were read by eye from the printed graph and carry an estimated reading uncertainty of ±0.05 in Cl and ±0.0005 in Cd.

Data Analysis

- **H1 was partially supported.** Maximum L/D increased with camber, with the NACA 0012, 2412, and 4412 reaching 75.6, 104.7, and 132.9, respectively, at Re = 1,000,000. However, the angle of maximum L/D did not consistently decrease, measuring 5.0°, 7.0°, and 9.0°, respectively, at Re = 100,000.
- **H2 was not supported.** Maximum L/D varied by less than 8% across airfoil thicknesses. Thicker airfoils produced slightly higher L/D, with about 27% more lift and 23% more drag for the NACA 0018 compared with the NACA 0012 at Re = 1,000,000.
- **H3 was supported.** Maximum L/D increased with Reynolds number for all five airfoils. The improvement from 100k to 500k was greater than from 500k to 1M. For example, the NACA 0012 gained 25.2 in maximum L/D from 100k to 500k, compared with only 13.8 from 500k to 1M.

Discussion

Validation and reliability

To assess XFOIL's accuracy, the NACA 2412 airfoil was simulated at a Reynolds number of 3.1 million and compared with wind-tunnel measurements from NACA Report 824. The results showed that XFOIL overestimated the maximum lift coefficient by approximately 11% and underestimated the minimum drag coefficient by roughly 16%. These differences are consistent with XFOIL's known limitations near stall conditions and its assumption of a perfectly smooth surface. Since all airfoils in this study were simulated using the same parameters, these errors are likely to affect each case in a similar way. Therefore, comparisons between airfoils are expected to be more reliable than the absolute values, which may be slightly optimistic.

Effect of thickness

The results showed that thicker airfoils generally produced more lift compared to the amount of drag they created, making them more efficient in some conditions. One interesting result was that at a Reynolds number of 100,000, the thinner NACA 0012 stalled at around 10° angle of attack, while the thicker NACA 0018 continued producing lift until around 14°. This suggests that thicker airfoils may perform better at higher angles before reaching stall.

Angle of maximum L/D at low Reynolds number

Another unexpected result was that cambered airfoils reached their maximum lift-to-drag ratio at higher angles of attack at low Reynolds numbers. This may happen because low Reynolds numbers create laminar separation bubbles, which increase drag at smaller angles. As the angle of attack increases, the airfoil produces more lift, allowing the lift-to-drag ratio to improve before eventually reaching its maximum.

Engineering significance

For a small drone flying at a Reynolds number of approximately 100,000, I would choose the 4412 because it had the highest lift-to-drag ratio in Table 1. A higher L/D ratio means the drone can produce the lift it needs with less drag, potentially improving flight time and battery efficiency.

Limitations

1. **Steady flow:** The simulation assumes airflow stays constant, but real conditions include wind gusts and turbulence. This can make actual performance different from the results.
2. **2D simulation:** XFOIL only analyzes a 2D airfoil and does not account for wingtip effects. This means the results may differ from a real 3D wing.
3. **Near stall:** XFOIL becomes less accurate when the airfoil approaches its stall angle. This makes predictions of lift and drag less reliable in that region.
4. **Limited validation:** Only one airfoil was used to check the accuracy of the simulation, and it was tested at a Reynolds number outside the main test range. This limits how confidently the results can be applied to other airfoils and conditions.

5. **Resolution:** The angle of attack was tested in 0.5° increments, so the exact angle that produces the highest lift-to-drag ratio may fall between two test points. This limits the precision of the results.

Conclusion

This study asked how changes in airfoil camber and thickness influence the maximum lift-to-drag ratio, and the angle of attack at which it occurs, at Reynolds numbers between 100,000 and 1,000,000. Increasing camber produced the largest gain: at Re = 1,000,000, $(L/D)_{max}$ rose from 75.6 for the NACA 0012 to 132.9 for the NACA 4412, although the angle of $(L/D)_{max}$ did not consistently decrease. Increasing thickness did not reduce $(L/D)_{max}$ as predicted; the thicker sections matched or slightly exceeded the NACA 0012, while also offering more internal volume for structure. As Reynolds number increased, inertial forces became more dominant over viscous forces, the boundary layer became thinner, and $(L/D)_{max}$ rose for every airfoil, with the largest gains between 100,000 and 500,000. Future work could test these airfoils in a wind tunnel at low Reynolds numbers, where no NACA data exist, and extend the analysis to 3D wings that include wingtip effects.

References

Abbott, I. H., & von Doenhoff, A. E. (1959). *Theory of wing sections: Including a summary of airfoil data*. Dover Publications.

Abbott, I. H., von Doenhoff, A. E., & Stivers, L. S., Jr. (1945). *Summary of airfoil data* (NACA Report No. 824). National Advisory Committee for Aeronautics.

Drela, M. (1989). XFOIL: An analysis and design system for low Reynolds number airfoils. In T. J. Mueller (Ed.), *Low Reynolds number aerodynamics* (Lecture Notes in Engineering, Vol. 54, pp. 1–12). Springer.

Lissaman, P. B. S. (1983). Low-Reynolds-number airfoils. *Annual Review of Fluid Mechanics, 15*, 223–239.

Mueller, T. J., & DeLaurier, J. D. (2003). Aerodynamics of small vehicles. *Annual Review of Fluid Mechanics, 35*, 89–111.

Appendices

Appendix A. Software and setup

Item	Version or setting
Computer	MacBook Air, Apple M4, macOS 15.7.8
XFOIL	6.99, installed with Homebrew from the `chrislupp/xfoil` package (patched for Apple Silicon)
X11 server	XQuartz (installed; XFOIL graphics were disabled with `PLOP` → `G`)
Python	3.9.6 (macOS system Python)
Plotting	matplotlib 3.9.4

Installation commands:

```
brew install --cask xquartz
brew tap chrislupp/xfoil
brew trust chrislupp/xfoil
brew install xfoil
python3 -m pip install --user matplotlib
```

Appendix B. XFOIL command sequence

The same commands were sent to XFOIL for every run. Example for the NACA 2412 at Re = 100,000:

```
PLOP
G

NACA 2412
PPAR
N 160


OPER
VISC 100000
MACH 0.0
```

```
VPAR
N 9

ITER 100
PACC
NACA2412_Re100000.txt

ASEQ 0.0 15.0 0.5
INIT
ASEQ 0.0 -5.0 -0.5
PACC

QUIT
```

The validation run used `ASEQ 0.0 20.0 0.5` for the upward sweep.

Appendix C. Scripts and data files

File	Contents
`run_sweeps.py`	Runs all XFOIL sweeps, removes duplicate rows, logs missing points, writes the CSV files
`plot_results.py`	Produces Figures 1–4 from the CSV files
`data/raw/NACAxxxx_Re<Re>.txt`	21 raw XFOIL polar files
`data/all_polars.csv`	Every converged point, with L/D
`data/summary.csv`	(L/D)max and its angle for each airfoil and Reynolds number
`data/validation_*.csv`	The NACA 2412 validation run at Re = 3,100,000
`data/run_log.txt`	Settings used and points that failed to converge

To reproduce the study: install the software in Appendix A, then run `python3 run_sweeps.py`, `python3 run_sweeps.py --validation-only`, and `python3 plot_results.py`.

Appendix D. Non-converged points

Airfoil	Reynolds number	Missing angles of attack
NACA 0012	200,000	6.5°
NACA 0012	500,000	−4.5°, 4.5°
NACA 2412	100,000	15.0°
NACA 2412	200,000	−2.5°
NACA 2412	1,000,000	−1.0°, 7.5°
NACA 4412	200,000	−5.0°, 11.0°
NACA 4412	1,000,000	−1.5°
NACA 0015	500,000	7.0°
NACA 0015	1,000,000	−2.5°, 2.5°, 7.0°
NACA 0018	1,000,000	−4.5°, 4.5°

All other runs converged at all 41 angles. Total: 16 of 820 points (2.0%).

Appendix E. Contribution statement

This project was completed by Aarnav Ravi with assistance from Claude, an AI assistant made by Anthropic, which acted as a research supervisor and technical assistant.

Part of the project	Aarnav	Claude
Research question and hypotheses	Approved the question; kept the hypotheses	Proposed the question; drafted H1–H3
Methodology	Approved the method; reviewed design choices	Designed the airfoil series and test conditions; wrote the Methodology section
Software setup	Installed and troubleshot XFOIL, XQuartz, and Python	Researched the Apple Silicon install route

Part of the project	Aarnav	Claude
Simulations	Ran all manual and scripted XFOIL runs	Wrote `run_sweeps.py`
Validation	Found NACA Report 824 and read the wind-tunnel values	Located the NACA 2412 page; computed percentage differences
Analysis and figures	Interpreted the hypotheses	Wrote `plot_results.py`; computed L/D; checked data quality
Writing: Introduction, Background fundamentals, Data Analysis, Limitations, Discussion	Wrote	Light edits for accuracy
Writing: Discussion opening (validation)	Rewrote from a sample	Drafted the sample
Writing: Prior work, Results, References, Abstract	Edited	Drafted
Writing: Conclusion	Wrote the research question and Reynolds-number explanation	Structured the section and added numbers